

Quantum Computing

- Week 6 (May 19-20, 2026)
- Topics:
 - Deutsch-Jozsa Problem
 - Entanglement
 - Pure states and mixed states

The Deutsch-Jozsa Problem

- Constant-vs-balanced problem
- Let $f: \{0,1\}^n \rightarrow \{0,1\}$ be a bit function such that it is in either two cases:
 - f is a *constant* function: $\forall x \in \{0,1\}^n, f(x)$ is always a constant (0 or 1)
 - f is a *balanced* function: $\sum_{x \in \{0,1\}^n} f(x) = 2^{n-1}$ (i.e., outputs 0 for half the inputs, and 1 for the other half)
- To decide whether f is constant or balanced, **how many times** must we evaluate f ?

Classical Computer

Worst-case: 2^{n-1} (Check half of inputs)

Probabilistic algorithm (Check l inputs):

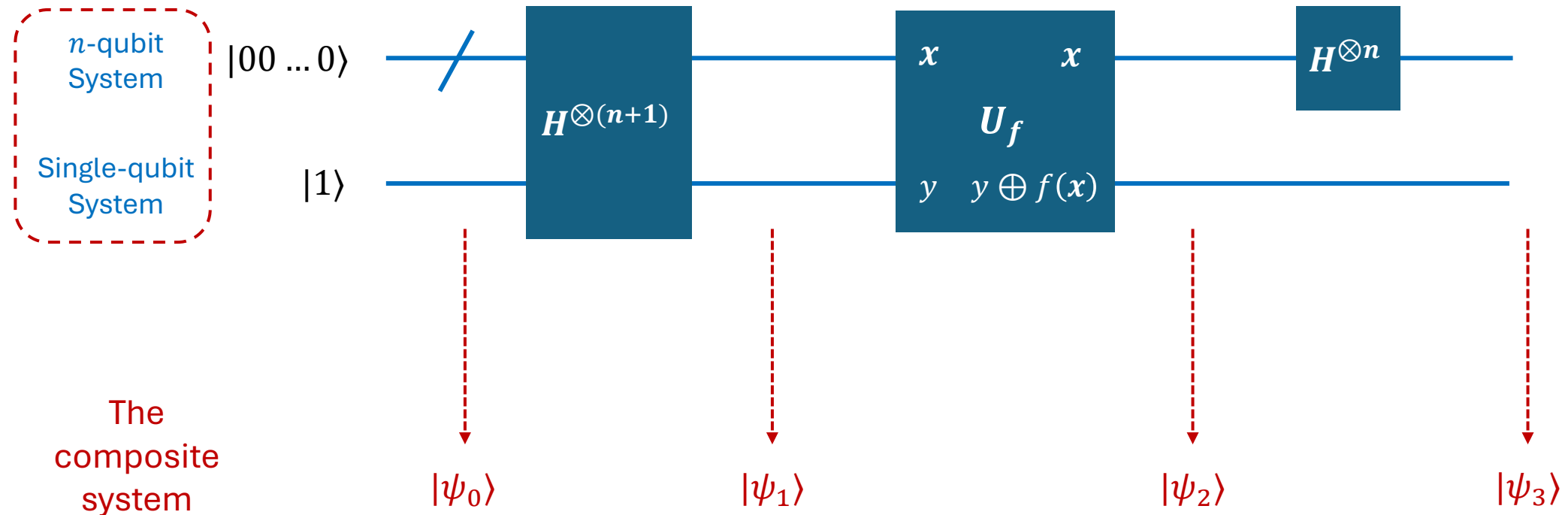
$l \ll 2^{n-1}$ times,

with a failure rate of $\frac{1}{2^l}$

Can we do better?

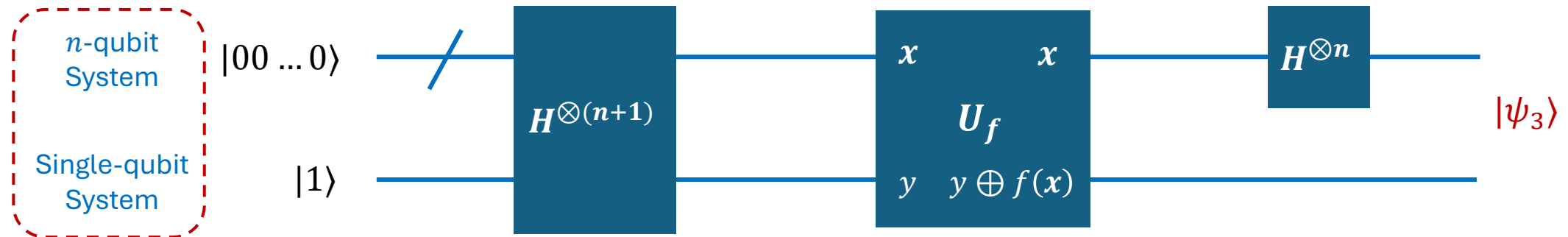
The Deutsch-Jozsa Algorithm

- Let $f: \{0,1\}^n \rightarrow \{0,1\}$ be a bit function...
- (Do it on the board)



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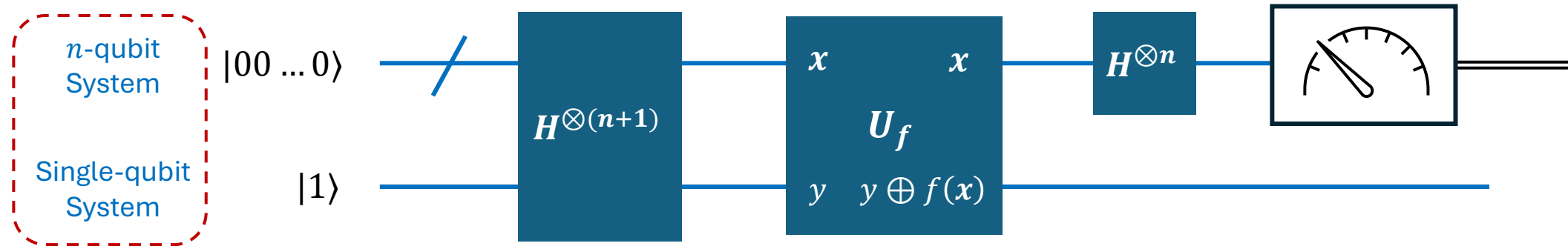


The
composite
system

$$|\psi_3\rangle = \left(\sum_{z \in \{0,1\}^n} \sum_{x \in \{0,1\}^n} \frac{(-1)^{x^T z + f(x)}}{2^n} |z\rangle \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

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Measure the
first n systems

?

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Measure the first
 n -qubit system $\mathbf{z}$

- What's the probability of $\mathbf{z} = 00 \dots 0$? What about $\mathbf{z} \neq 00 \dots 0$?

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Supremacy

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Quantum
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But wait...What's the
practical application of the
Deutsch-Jozsa problem?

Quantum-Classical Separation Problems

- The Deutsch–Jozsa problem has no known practical application
- It is an early example of quantum supremacy, illustrating (or suggesting) the theoretical **separation between quantum and classical computation** (e.g., BPP vs BQP)

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- Similar quantum-classical separation problems:
 - Simon's problem (exercise or homework, TBD)

Quantum-Classical Separation Problems

- The Deutsch–Jozsa problem has no known practical application
- It is an early example of quantum supremacy, illustrating (or suggesting) the theoretical separation between quantum and classical computation (e.g., BPP vs BQP)
- Similar quantum-classical separation problems:
 - Simon's problem
- (Potential) separation problems that have **practical use**:
 - Hidden subgroup problem (Discrete logarithm, Factoring): Shor's algorithm
- No quantum supremacy, but **quantum acceleration**
 - Unstructured search problem: Grover's search algorithm

Postulates of Quantum Computing

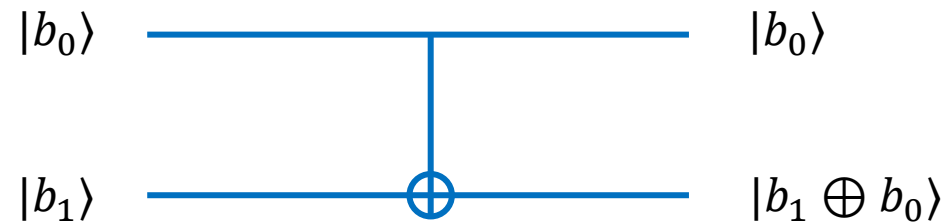
- Postulate 1: State space
- Postulate 2: Evolution and unitary transformation
- Postulate 3: Quantum Measurement
 - Projective measurement
- Postulate 4: Composite system

Postulates of Quantum Computing

- Postulate 1: State space (**isolated systems**)
- Postulate 2: Evolution and unitary transformation (**closed systems**)
- Postulate 3: Quantum Measurement
 - Projective measurement
- Postulate 4: Composite system

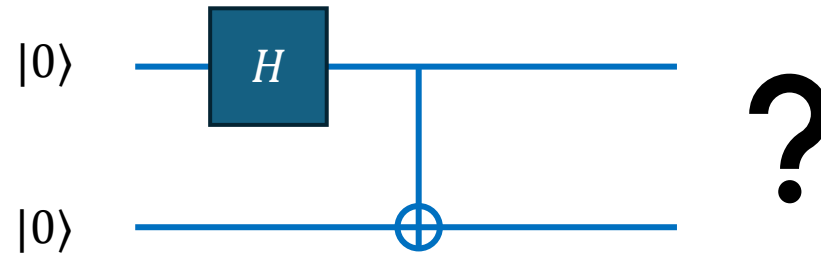
Controlled NOT Gate

- CNOT: If $b_0 = 0$, output b_1 ; Else, output $1 \oplus b_1$ (i.e., flip b_1 if $b_0 = 1$)



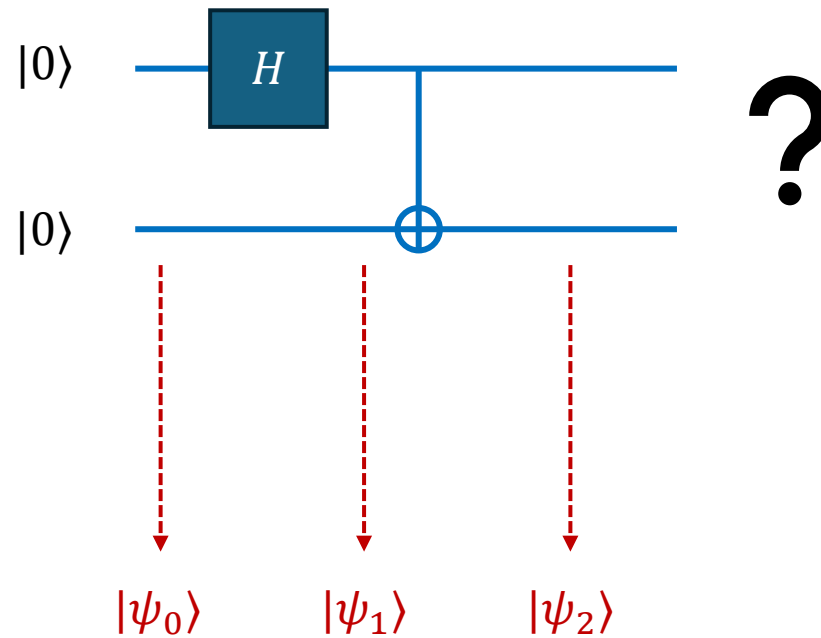
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Controlled NOT Gate



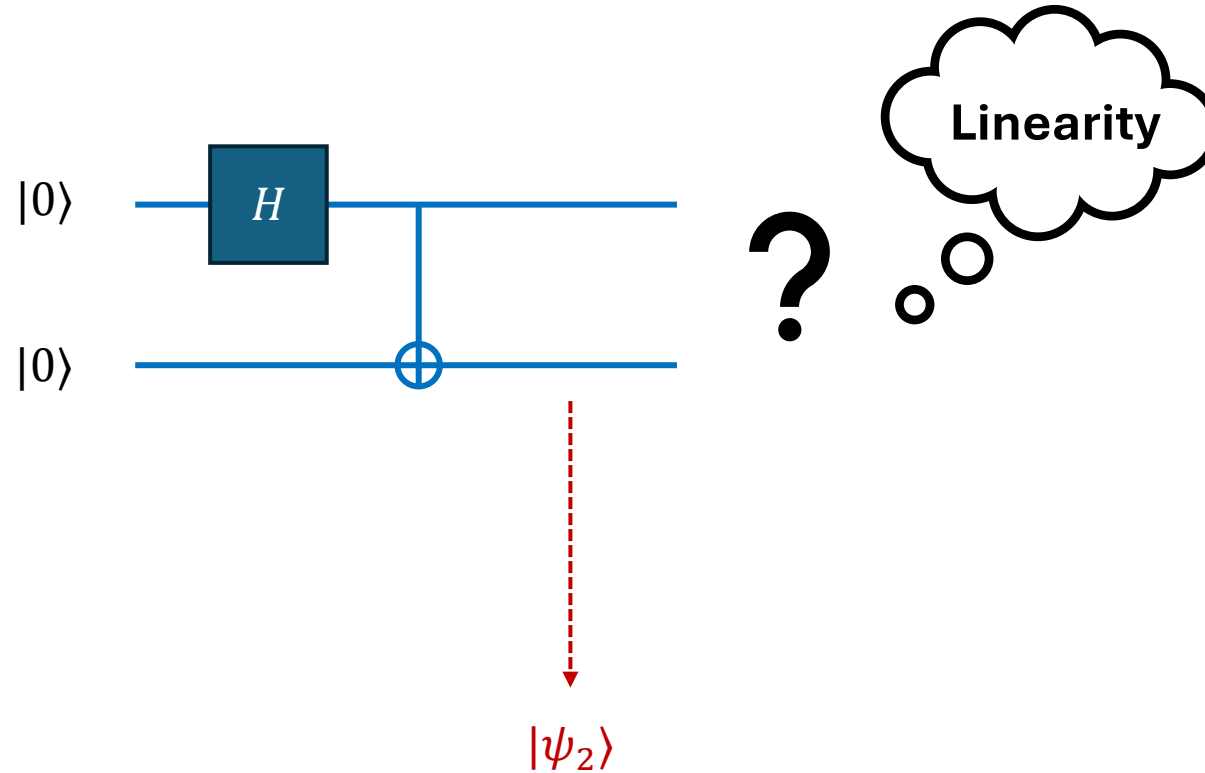
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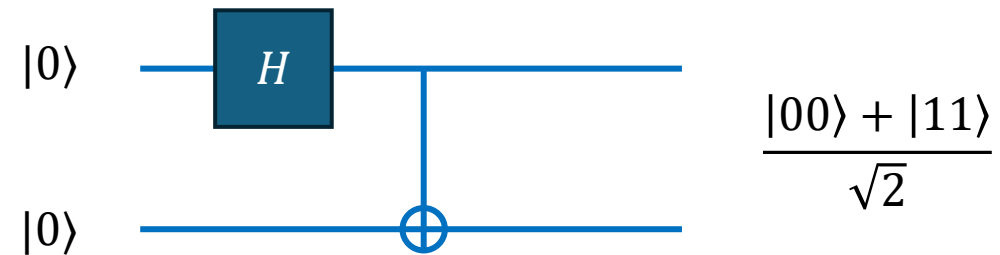
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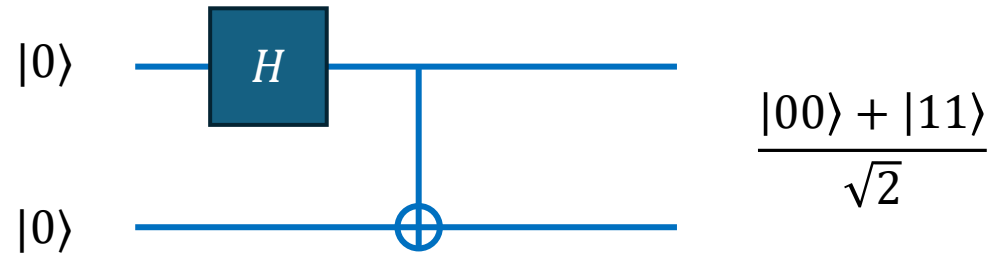
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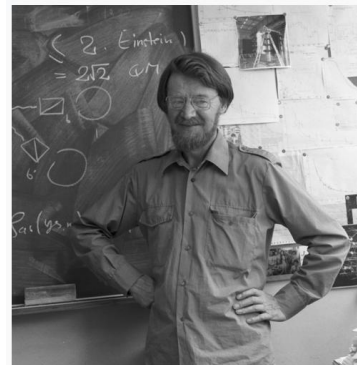


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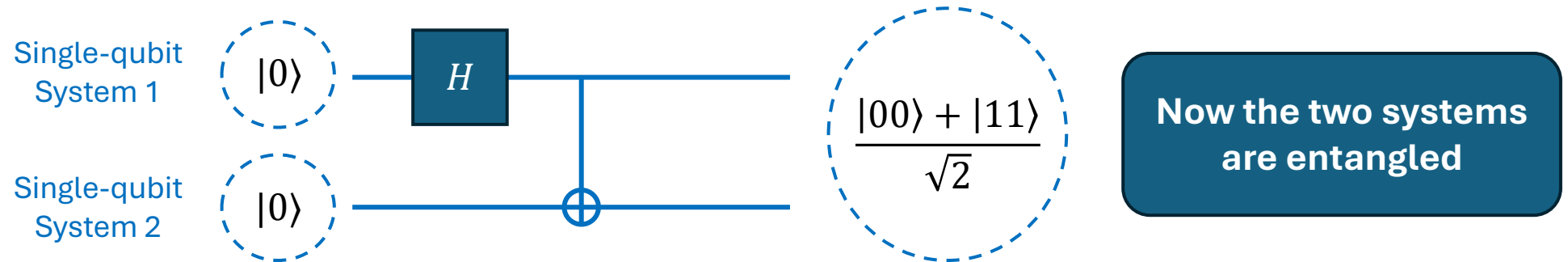
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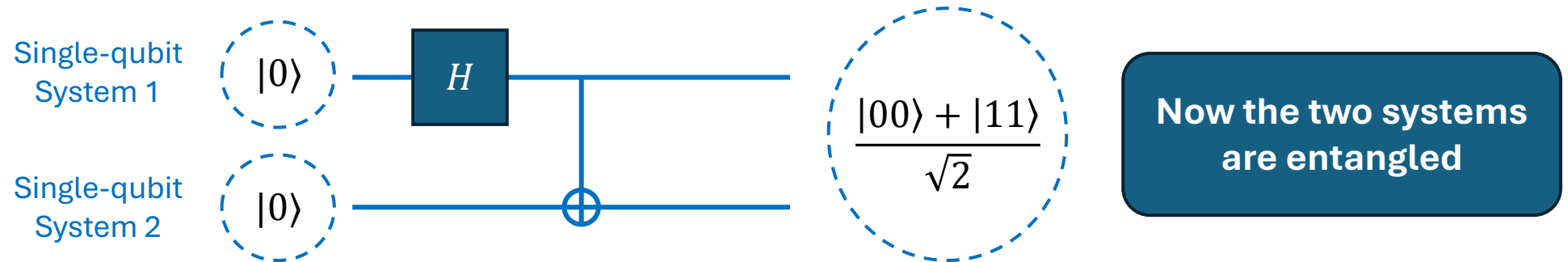
John Stewart Bell
(source: Wikipedia)

Bell state:
Impossible to be split into
a tensor product of two states
 $|\varphi_1\rangle \otimes |\varphi_2\rangle$

Quantum Entanglement

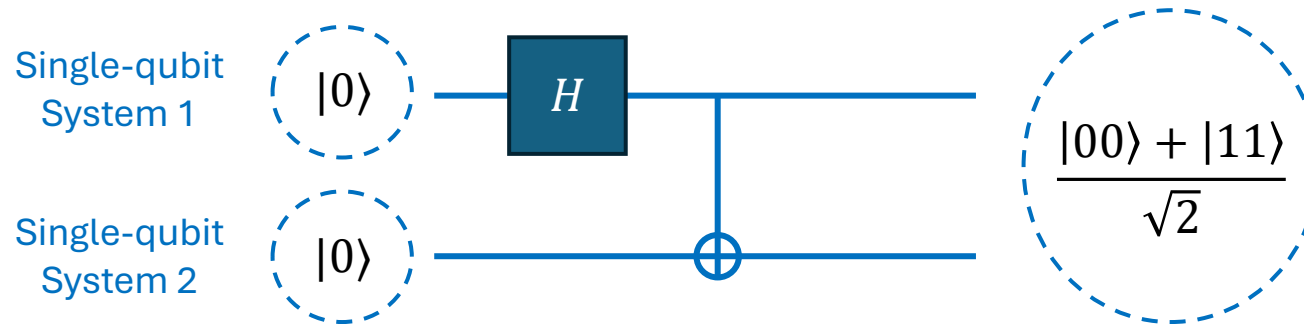


Quantum Entanglement



Pure state: Can be described by a state vector
Mixed state: Cannot ...

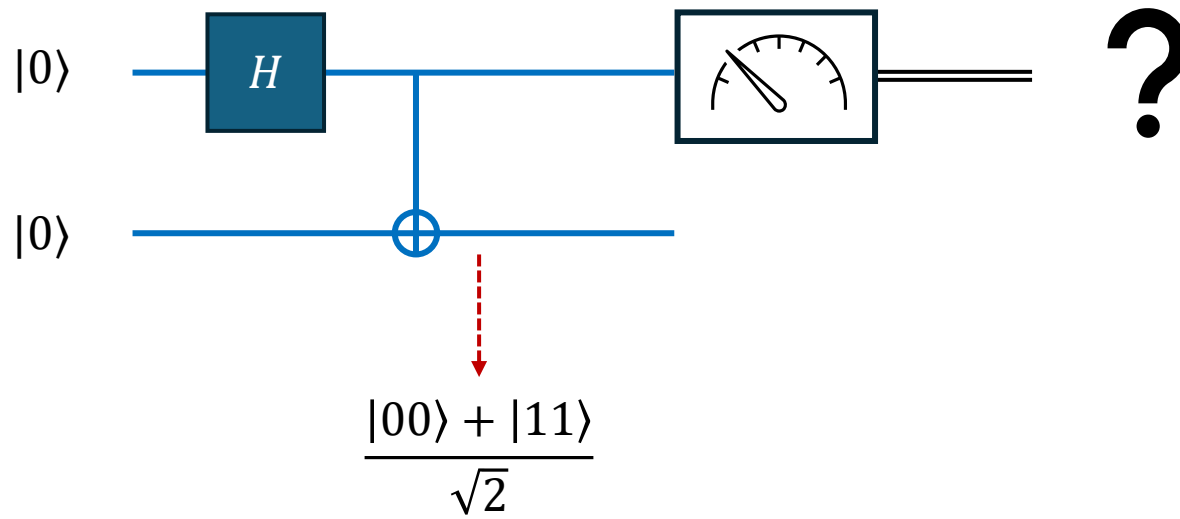
Quantum Entanglement



Small Exercise: (pure or mixed)

1. The initial state of system 1 is ____.
2. The states of system 1 and 2 (after H and CNOT) are ____.
3. The state of the total system (after H and CNOT) is ____.

Partial Measurement on Entangled States



Partial Measurement on Entangled States

- Formalizing Partial Measurement
 - Let's focus on the computational basis
 - General measurement: $\{M_m\}_m \rightarrow \{M_m \otimes I\}_m$
 - Projective measurement: $\mathbf{M} = \sum_m m \mathbf{P}_m \rightarrow \mathbf{M} \otimes \mathbf{I} = (\sum_m m \mathbf{P}_m) \otimes \mathbf{I} = \sum_m m (\mathbf{P}_m \otimes \mathbf{I})$

Partial Measurement on Entangled States

- **Spectral decomposition (simplified):**

Any Hermitian operator $\mathbf{M}$ (i.e., $\mathbf{M} = \mathbf{M}^\dagger$) can be written as:

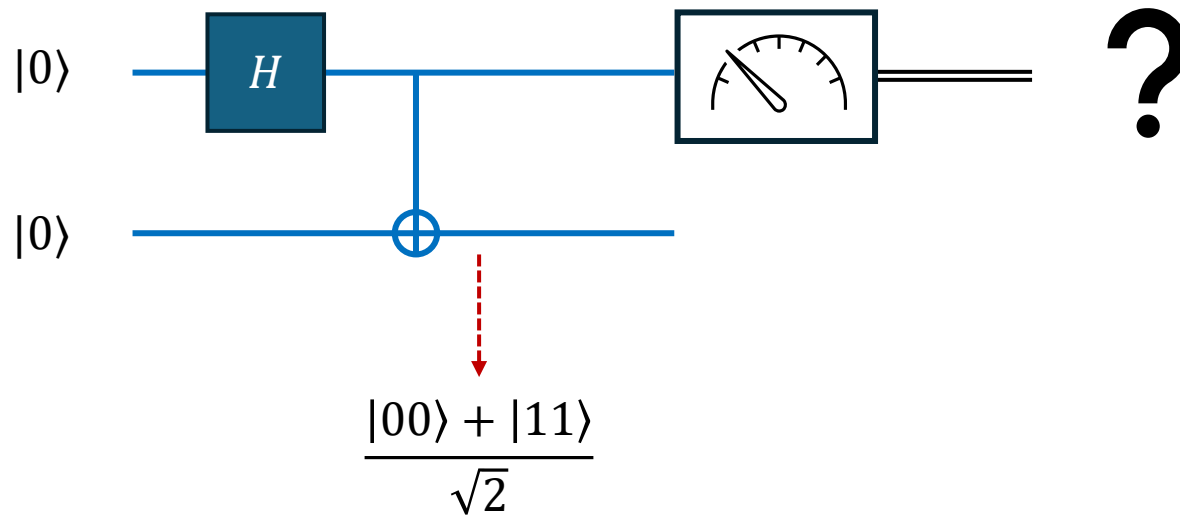
$$\mathbf{M} = \sum_{\lambda} \lambda \mathbf{P}_{\lambda}$$

- λ represents an eigenvalue of $\mathbf{M}$ (also indexes the projectors)
- $\mathbf{P}_{\lambda}$ represents the projector onto the λ eigenspace, $\mathbf{P}_{\lambda}^2 = \mathbf{P}_{\lambda}$
- $\mathbf{P}_{\lambda}$ itself is also Hermitian, i.e., $\mathbf{P}_{\lambda} = \mathbf{P}_{\lambda}^\dagger$
- $\sum_{\lambda} \mathbf{P}_{\lambda} = I$
- $\mathbf{P}_{\lambda}^\dagger \mathbf{P}_{\lambda'} = \delta_{\lambda, \lambda'} (= 1 \text{ if } \lambda = \lambda', \text{ otherwise } = 0)$

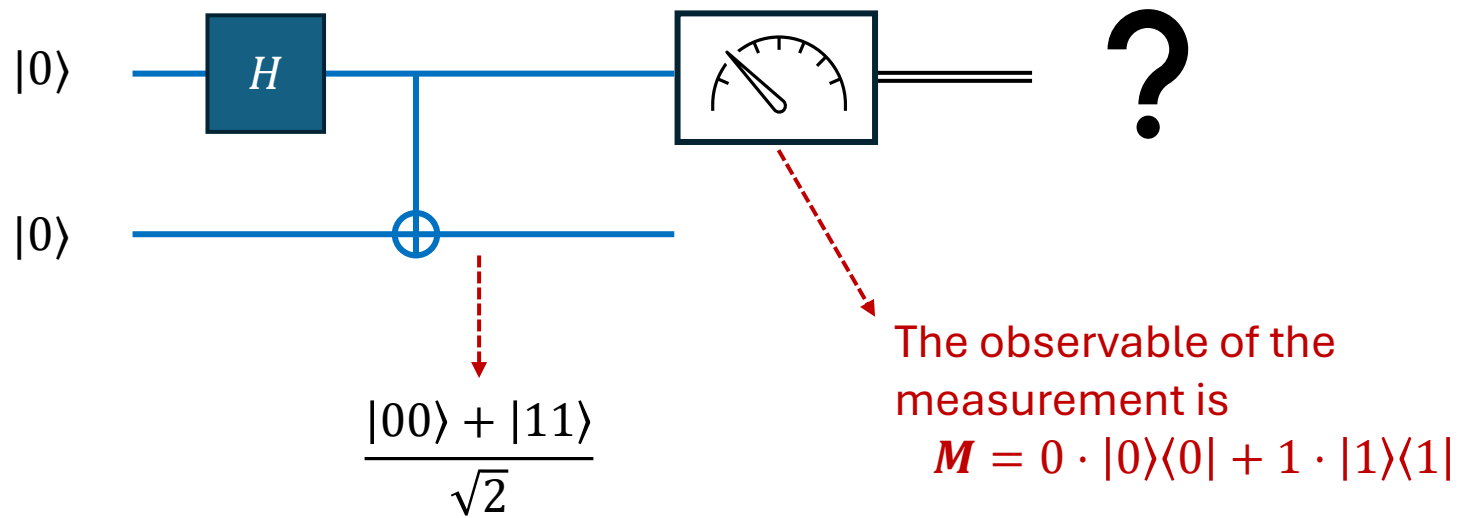
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- Important notes:
 - $\{M_m \otimes I\}_m$ still satisfies the completeness equation
 - $\mathbf{M} \otimes \mathbf{I}$ is still an observable
- Example (Exercise):
 - Partial measurement on the state $\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$

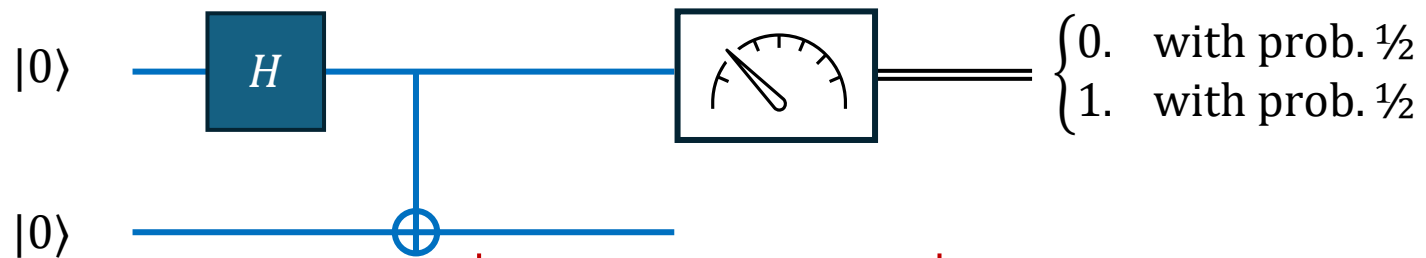
Partial Measurement on Entangled States



Partial Measurement on Entangled States



Partial Measurement on Entangled States

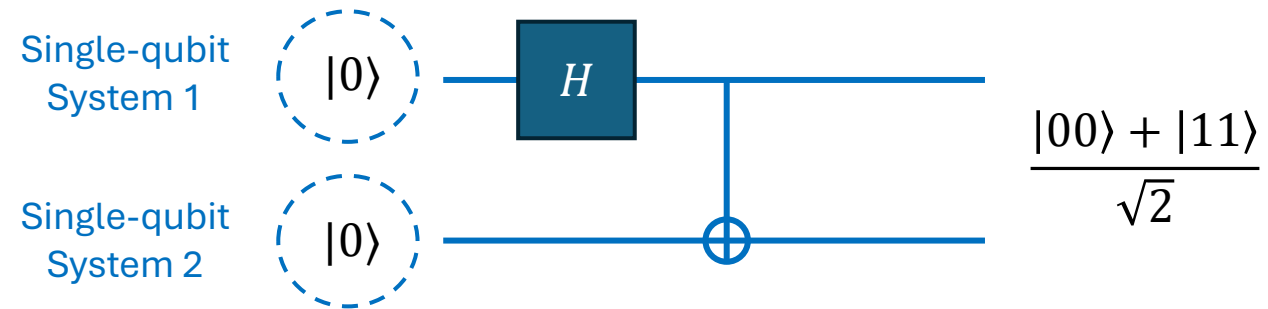


$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

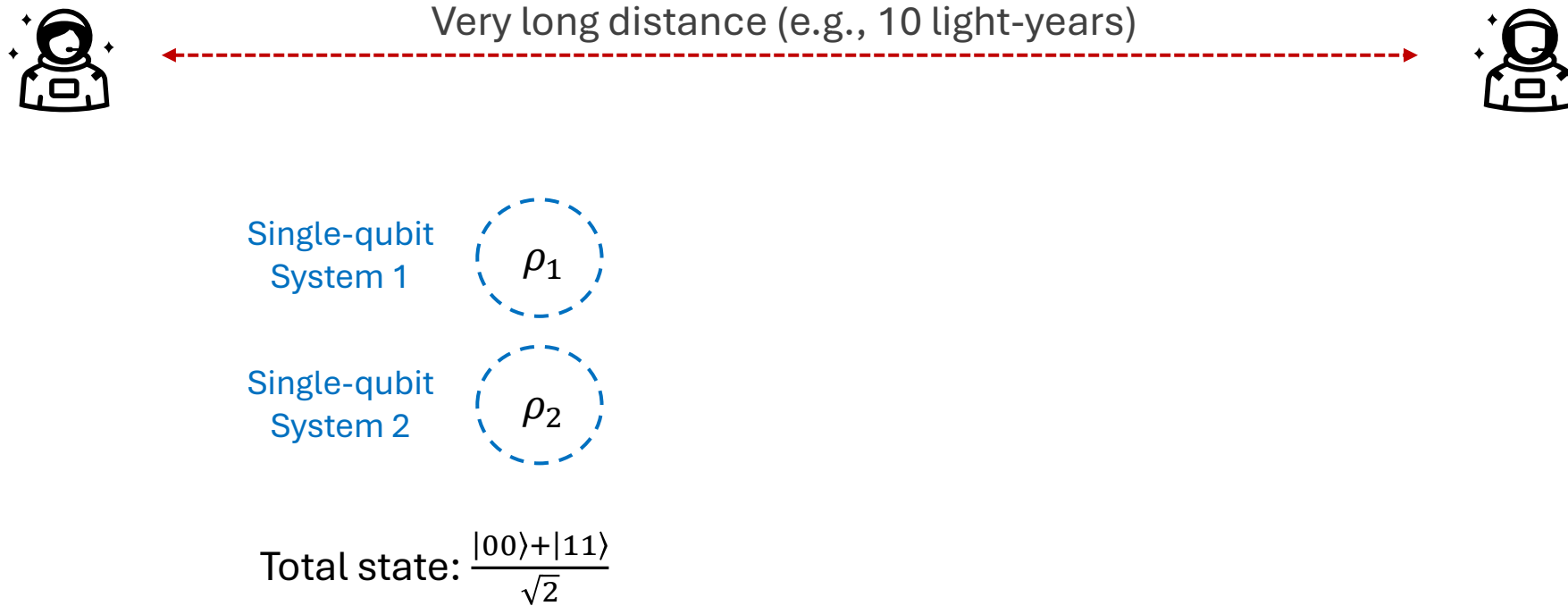
State after measurement $\begin{cases} |00\rangle & \text{if the measurement outcome is 0} \\ |11\rangle & \text{if the measurement outcome is 1} \end{cases}$

For entangled states,
partial measurement
leads to global collapse

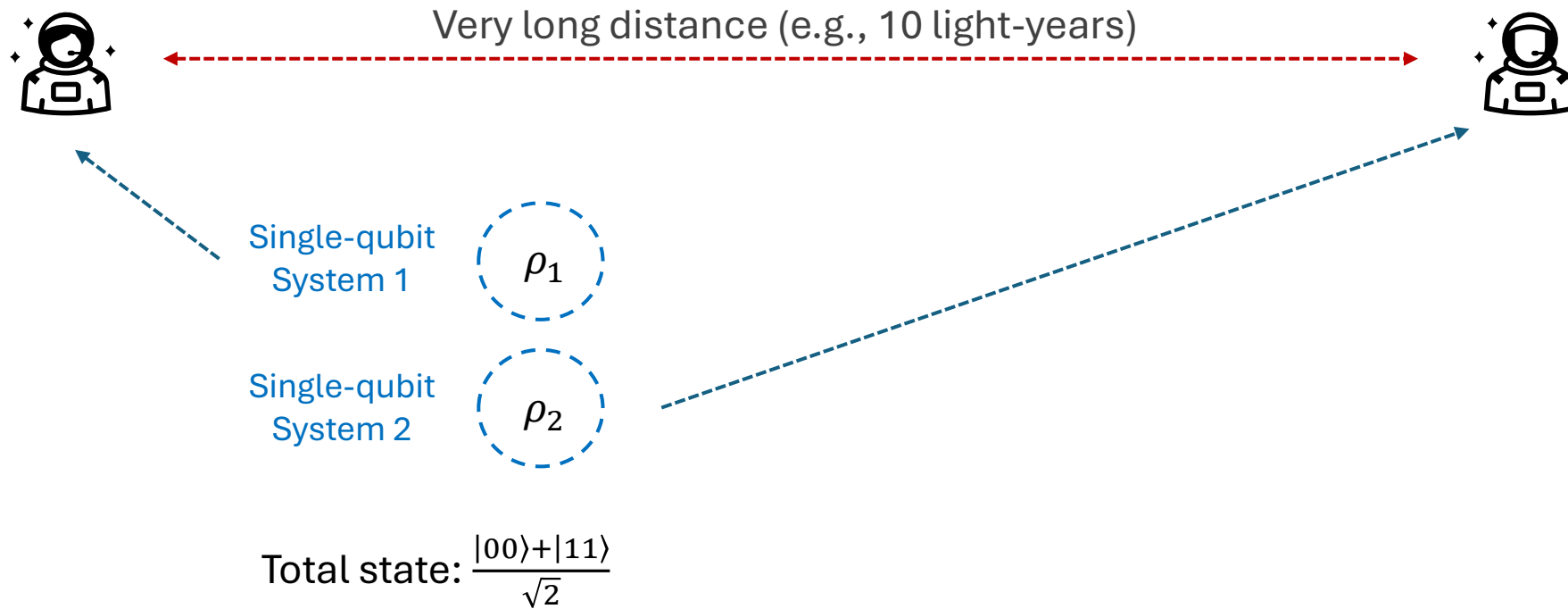
Action at a Distance (Fernwirkung)



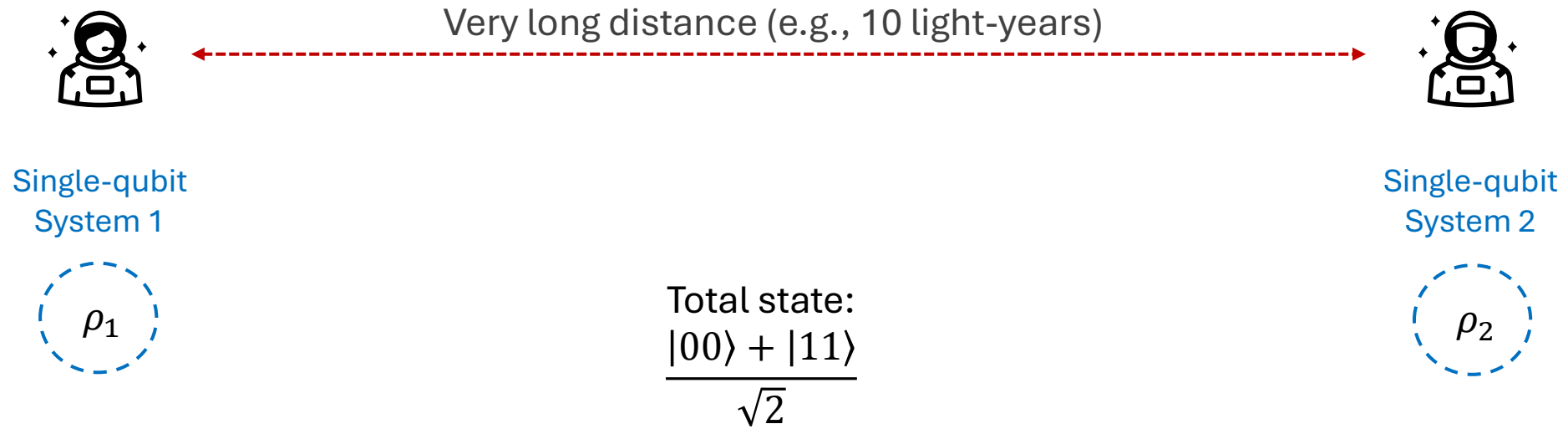
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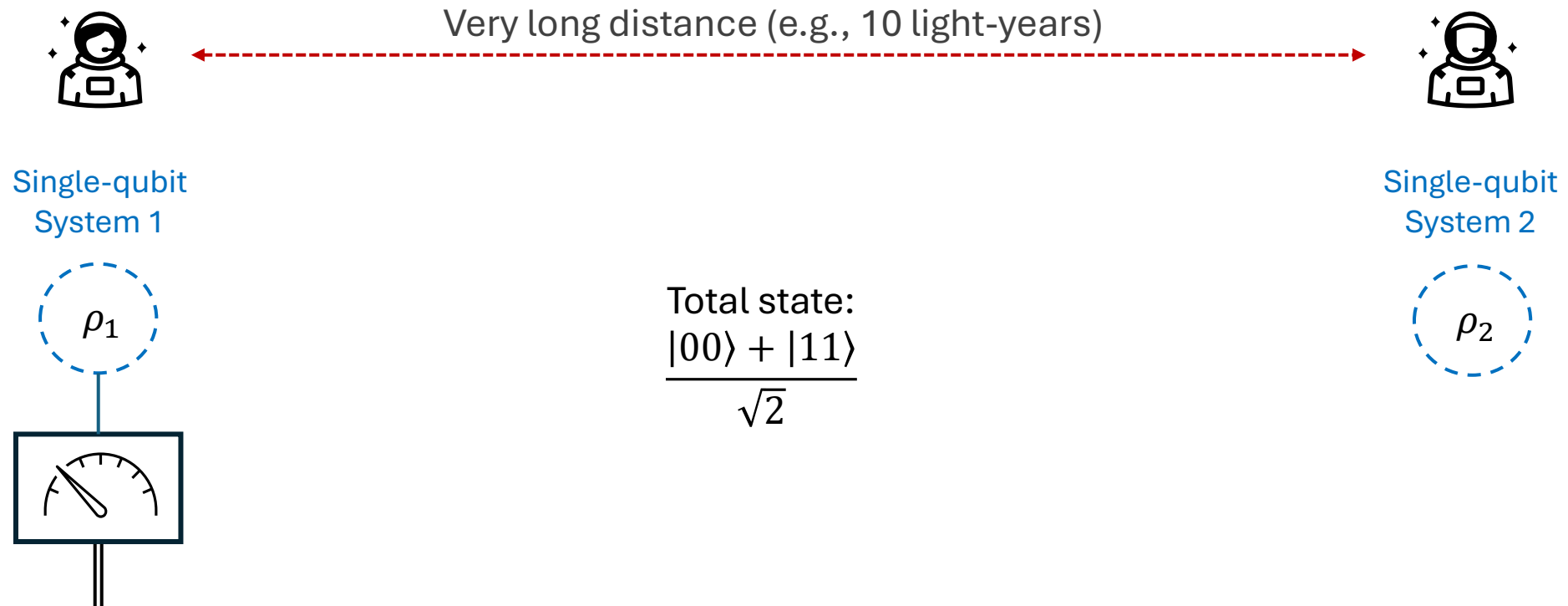
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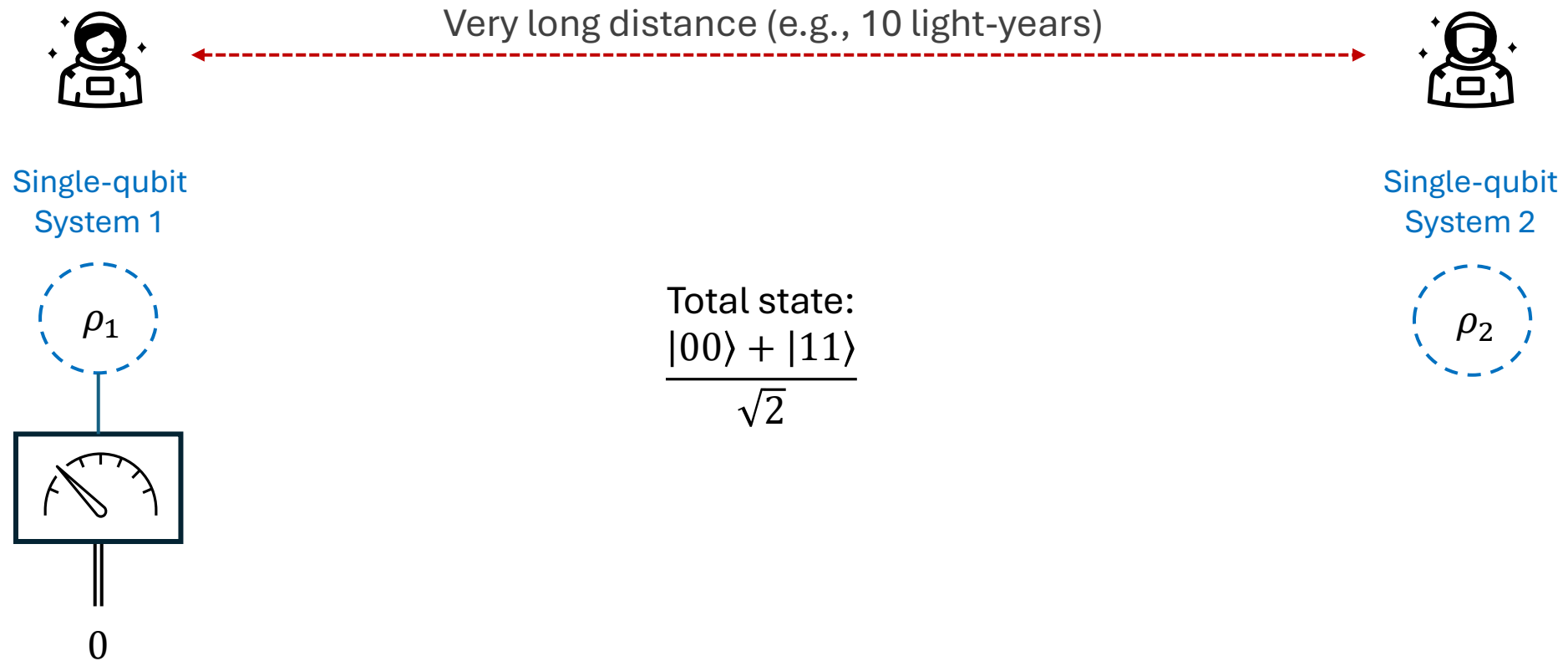
Action at a Distance



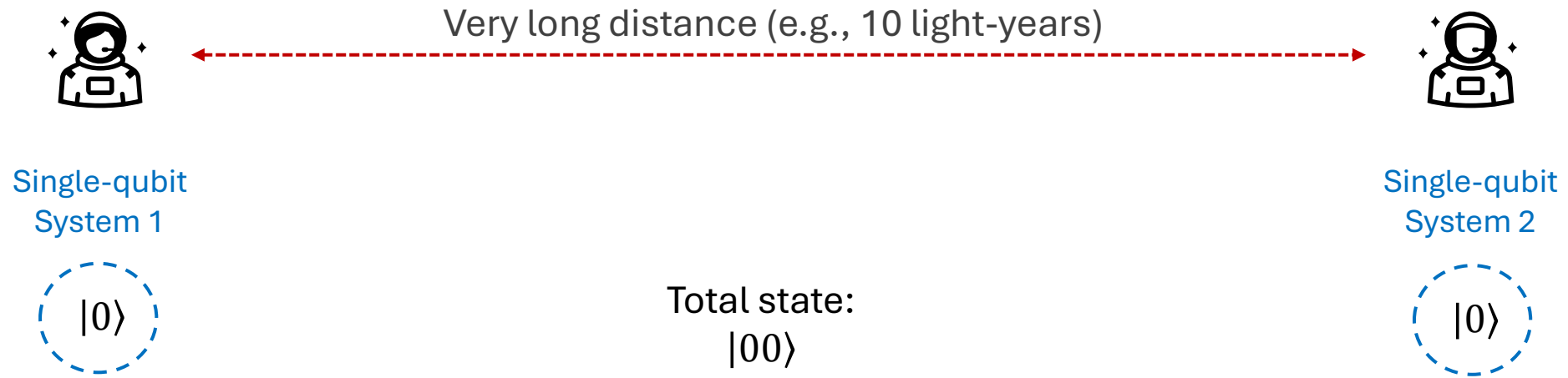
Action at a Distance



Action at a Distance

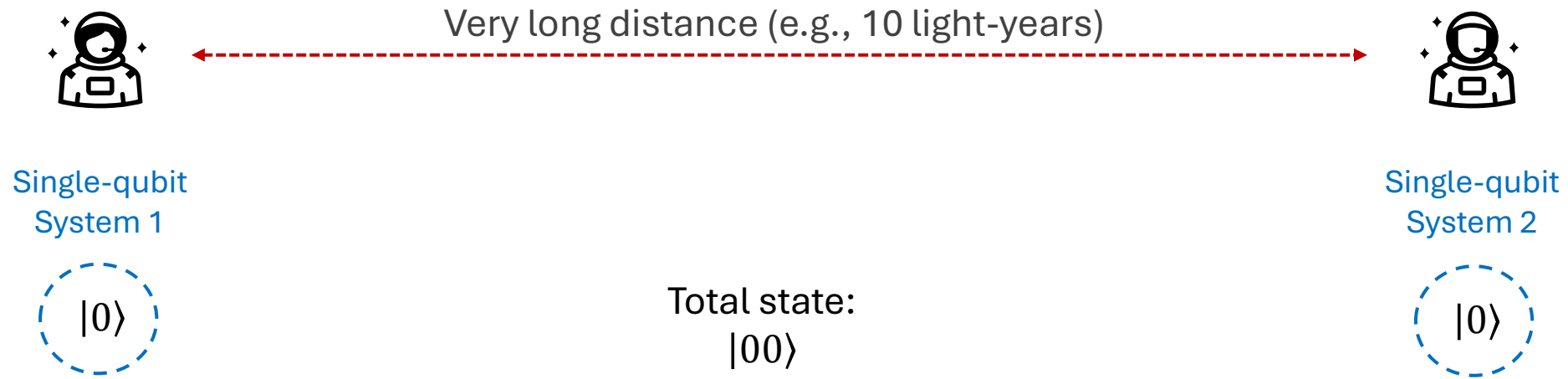


Action at a Distance



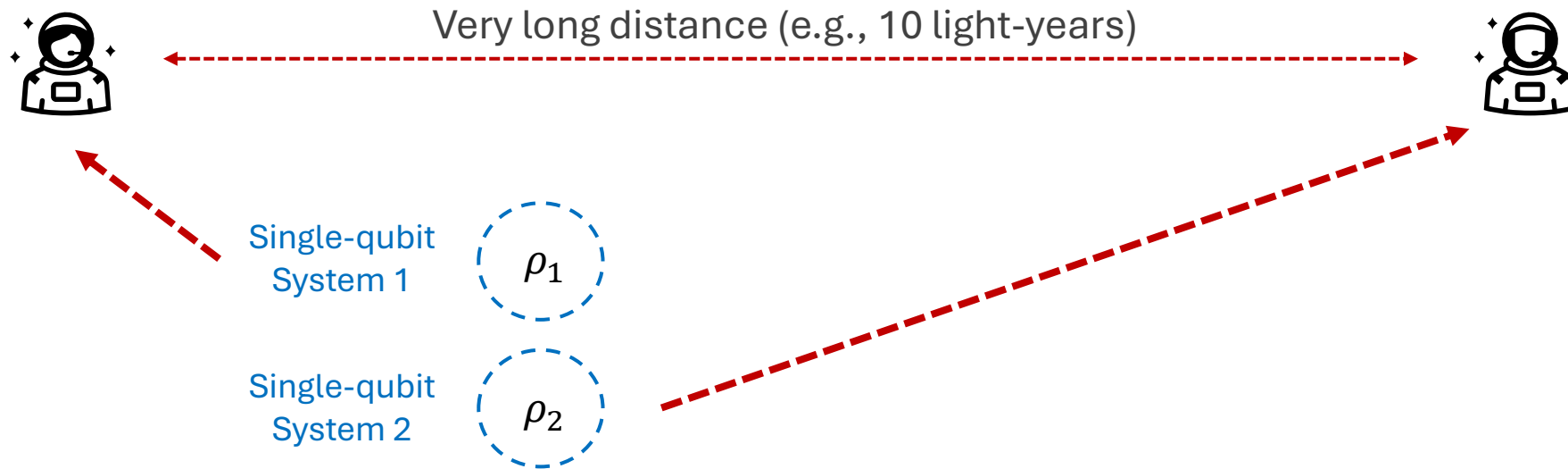
- “spukhafte Fernwirkung”

Action at a Distance



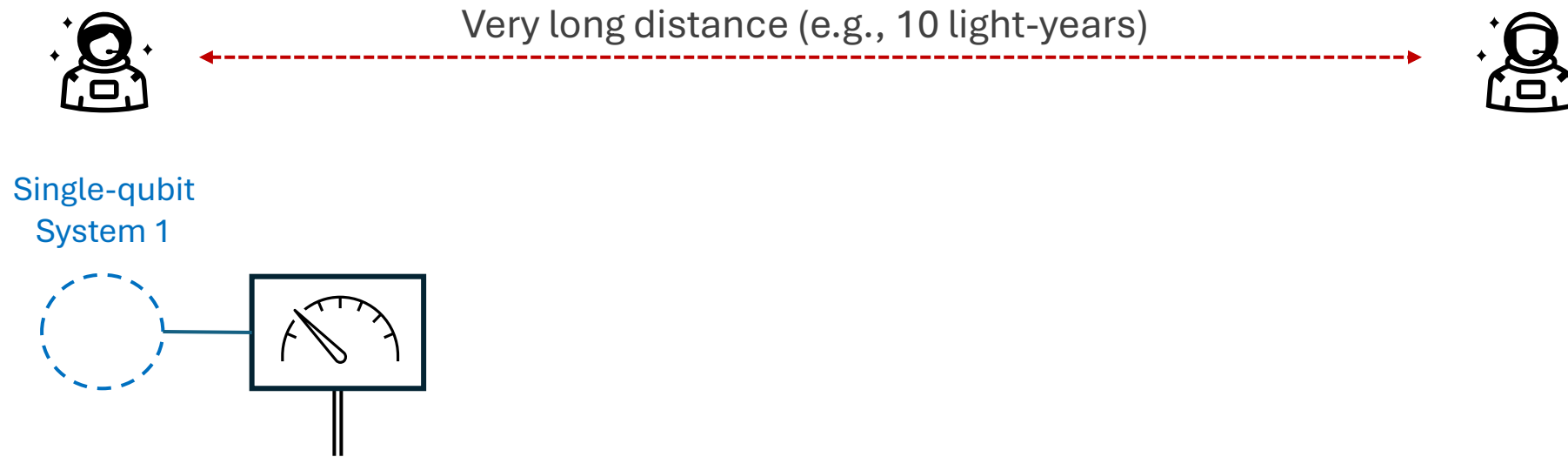
- “spukhafte Fernwirkung”
- A quick question: Is it a faster-than-light **communication**?

Action at a Distance



- “spukhafte Fernwirkung”
- A quick question: Is it a faster-than-light **communication**? **No**
 - They require some pre-shared entangled states

Action at a Distance



- “spukhafte Fernwirkung”
- A quick question: Is it a faster-than-light **communication**? **No**
 - They require some pre-shared entangled states
 - We cannot control the measurement output (i.e., random bit)
 - The receiver has no idea if its system has been measured

Next Week

- Superdense coding (...one qubit “encodes” two classical bits...)
- Quantum teleportation (...transfer one qubit via sending two classical bits...)